Class 3

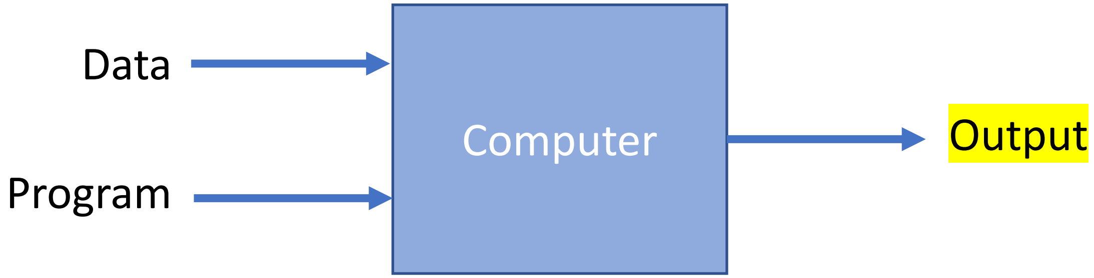
CSX4609/ITX4609 AI for Business

Machine Learning

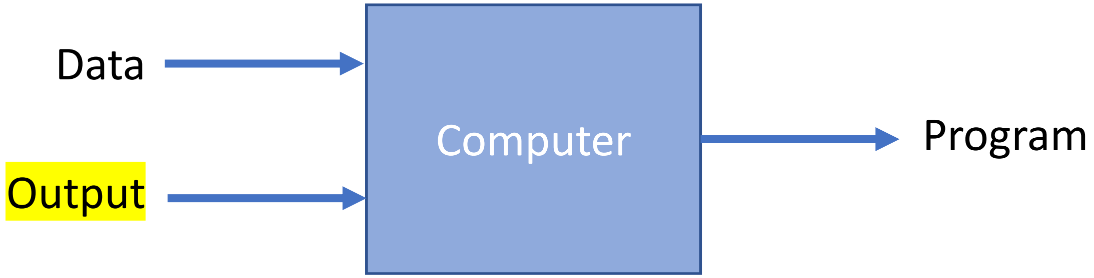
SEMESTER 1/2025

What is Machine Learning?

Traditional Programming



Machine Learning



What is Machine Learning?

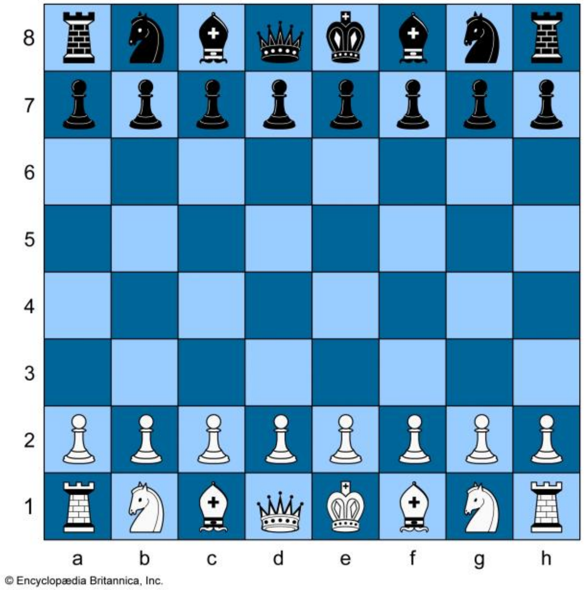
How do we learn?

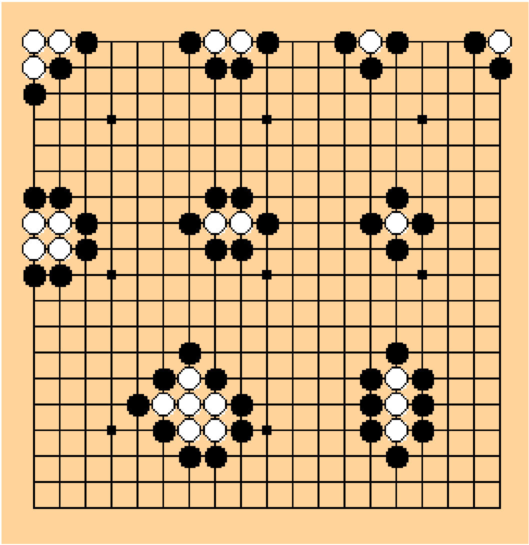
- Declarative knowledge (knowing **what**)
- Imperative knowledge (knowing **how-to-do**)

How should a computer learn?

What is Machine Learning?

Arthur Samuel (1959).
Machine learning: Field of study that **gives computers the ability to learn WITHOUT being explicitly programmed**.





What is Machine Learning?

Tom Mitchell (1998)

Well-posed learning problem is one that has **a clear task**, **a defined performance measure**, and **available experience** to learn from, ensuring the learning process is feasible and effective.

A computer program is said to LEARN from experience E with respect to some task T and some performance measure P, IF its performance on T, as measured by P, IMPROVES with experience E.

What is Machine Learning?

Suppose your email program watches which emails you do or do not mark as spam and based on that learns how to better filter spam. What is the task T in this setting?

a) Classifying emails as spam or not spam.
b) Watching you label emails as spam or not spam.
c) The number (or fraction) of emails correctly classified as spam/not spam
d) None or the above – this is not a machine learning problem.

What is Machine Learning?

Machine learning algorithms are largely used to

- predict
- classify
- cluster

Examples of Well-posed Learning Problems:

- Spam filtering
- Handwritten digit recognition
- Playing checkers
- Object recognition in images (e.g., identifying fruits or faces)
- Machine translation
- Driving a robot car

Machine Learning Algorithms

Supervised learning

teach the computer how to do something

Unsupervised learning

to let it learn by itself

Others:

reinforcement learning, recommender systems.

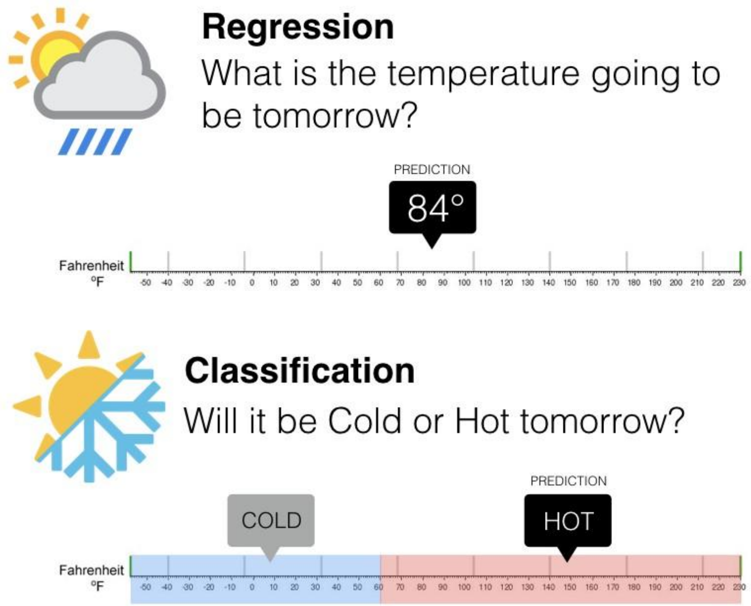
Supervised Learning

Regression - Continuous Variables

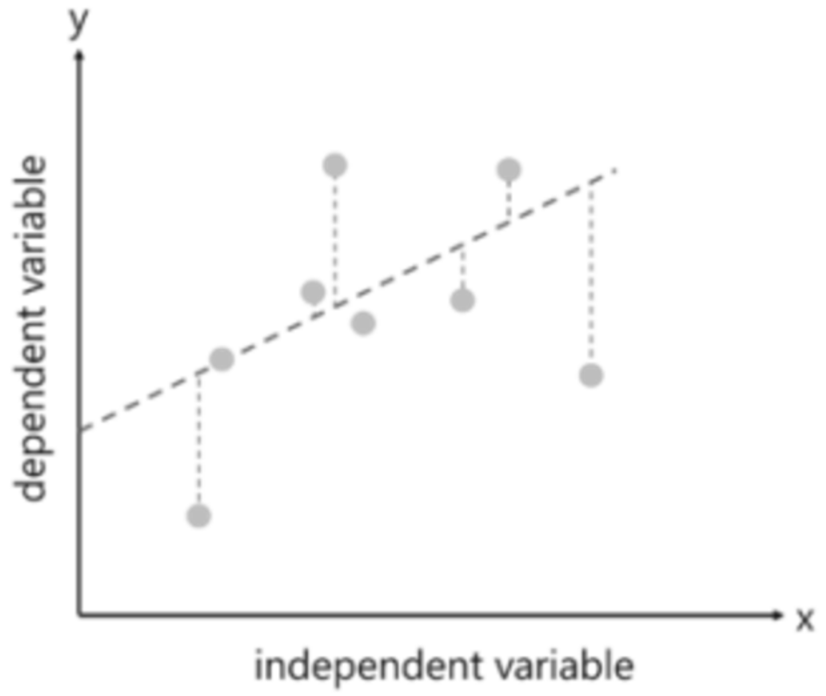
➢ Linear Regression

Classification - Categorical Variable

➢ k-Nearest Neighbors (k-NN)



Linear Regression



linear relationship between the 2 variables, Input (X) and Output (Y), of the data it has learnt from.

Linear Regression – An Example

- Housing price prediction

Number of rooms	Price
1	155
2	197
3	244
4	?
5	356
6	407
7	448

Ref: https://livebook.manning.com/book/grokking-machine-learning/3-2-the-solution-building-a-regression-model-for-housing-prices/v-4/47

Linear Regression – An Example (*Cont.*)

- Housing price prediction

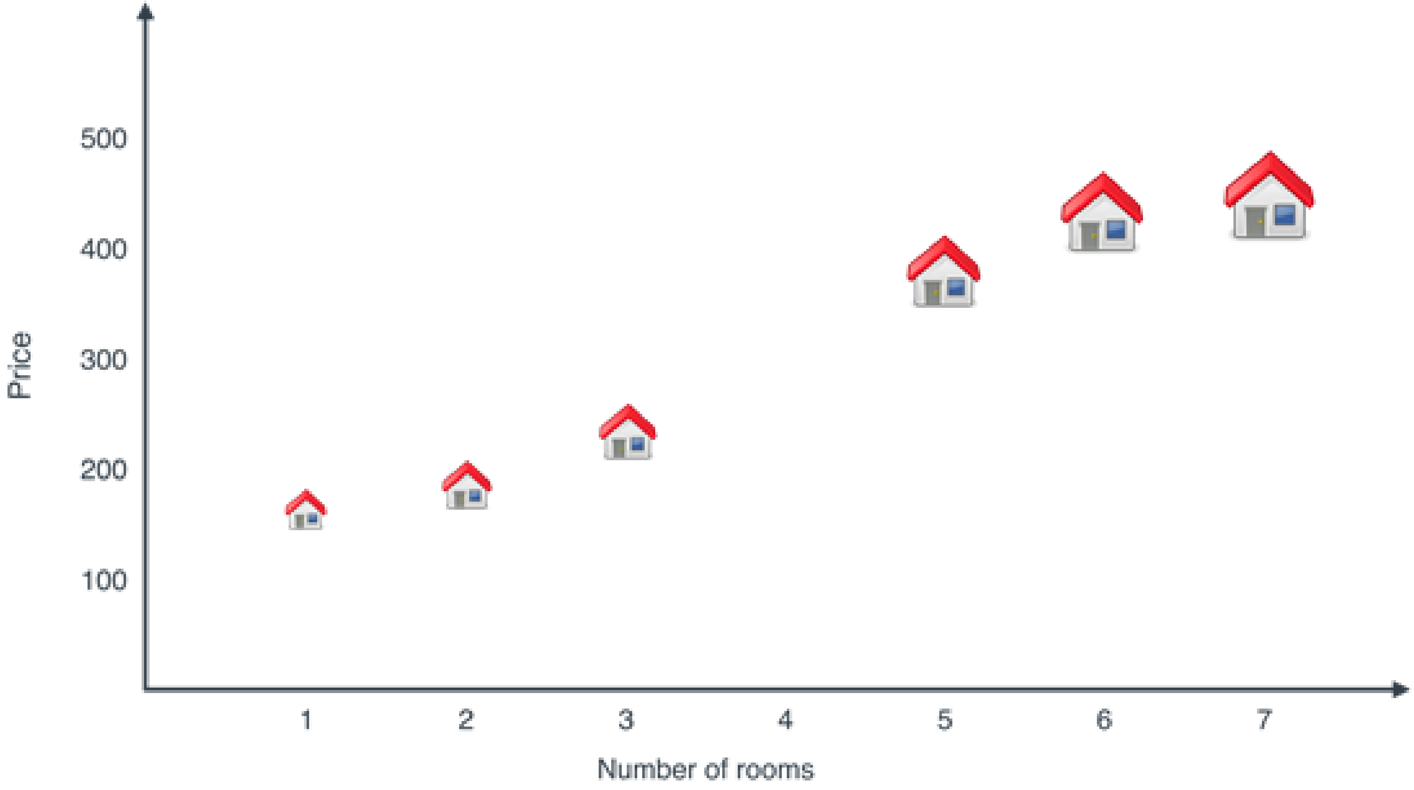
Number of rooms	Price
1	155
2	197
3	244
4	?
5	356
6	407
7	448

Price = 100 + 50 * (Number of rooms)

Ref: https://livebook.manning.com/book/grokking-machine-learning/3-2-the-solution-building-a-regression-model-for-housing-prices/v-4/47

Linear Regression – An Example (*Cont.*)

- Housing price prediction



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Linear Regression – An Example (*Cont.*)

- Housing price prediction

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Linear Regression

- Tomorrow’s stock market price
- The age of a viewer on YouTube.
- The amount of prostate specific antigen (PSA)
- The temperature at any location inside a building

Classification – KNN Classification Algorithm

- Is a simple algorithm that **stores all available cases and classifies new cases** based on a similarity measure (e.g., distance functions)
- Used in statistical estimation and pattern recognition already in the beginning of 1970's as a non-parametric technique.

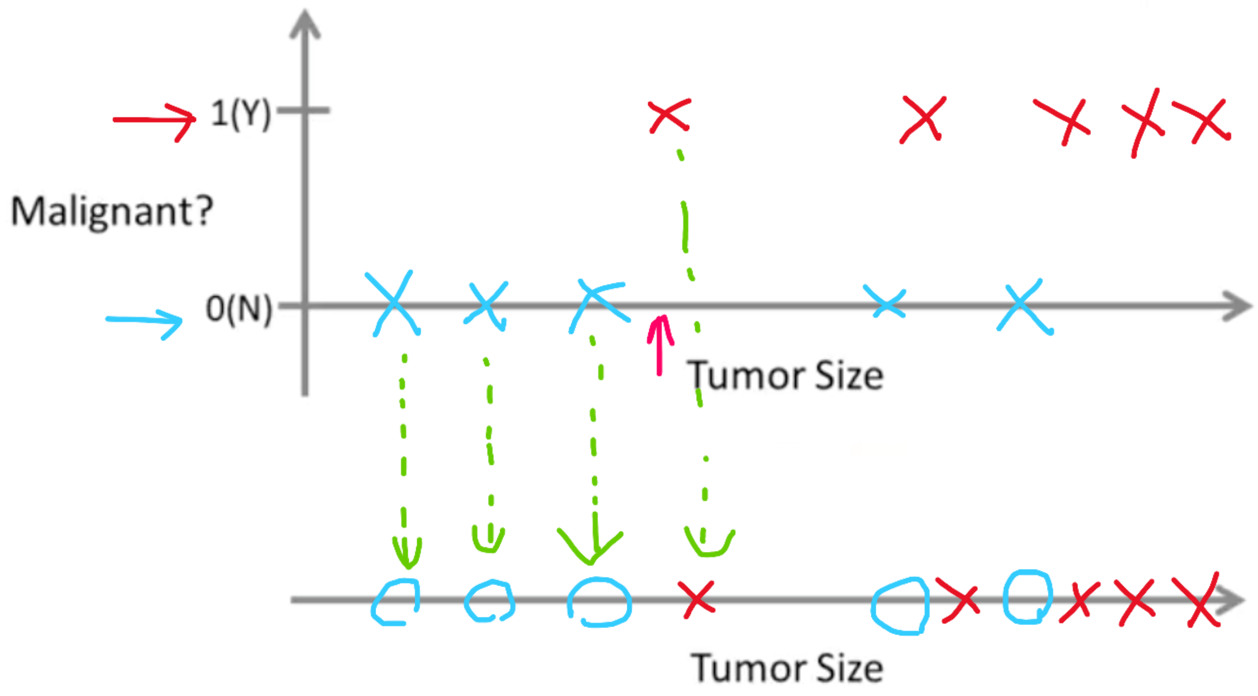
Classification

Breast cancer

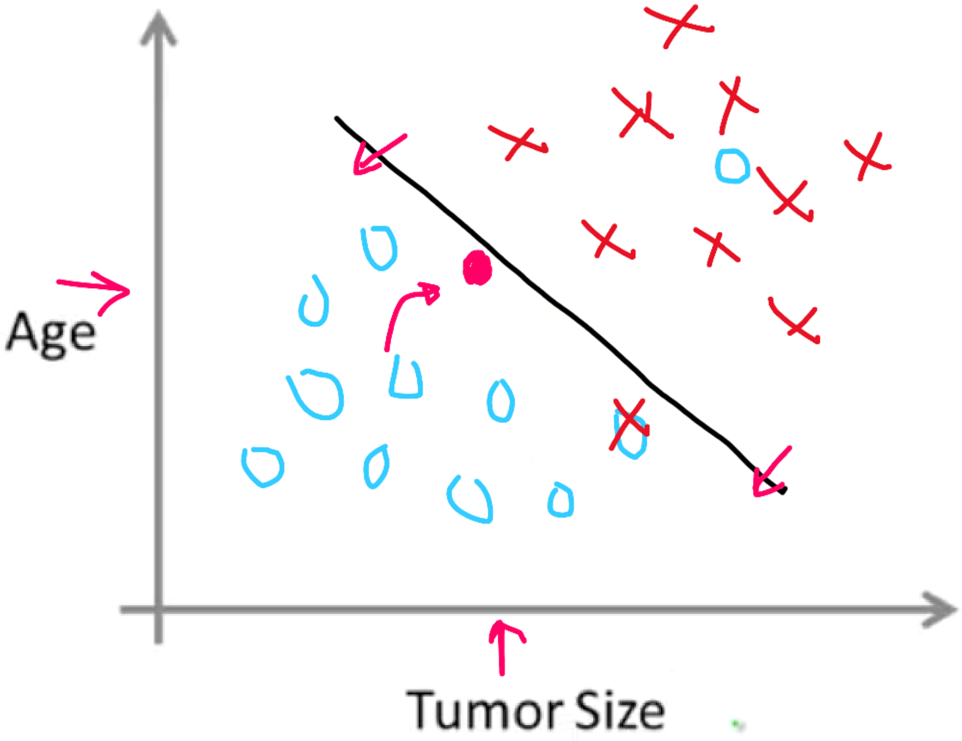
(malignant, benign)

Calssification

Discrete valued output (0 or 1)



Classification



- Clump Thickness
- Uniformity of Cell Size
- Uniformity of Cell Shape

…

You're running a company, and you want to develop learning algorithms to address each of two problems.

Problem 1: You have a large inventory of identical items. You want to predict how many of these items will sell over the next 3 months.
Problem 2: You'd like software to examine individual customer accounts, and for each account decide if it has been hacked/compromised.

Should you treat these as classification or as regression problems?

- A Treat both as classification problems.
- B Treat problem 1 as a classification problem, problem 2 as a regression problem.
- C Treat problem 1 as a regression problem, problem 2 as a classification problem.
- D Treat both as regression problems.

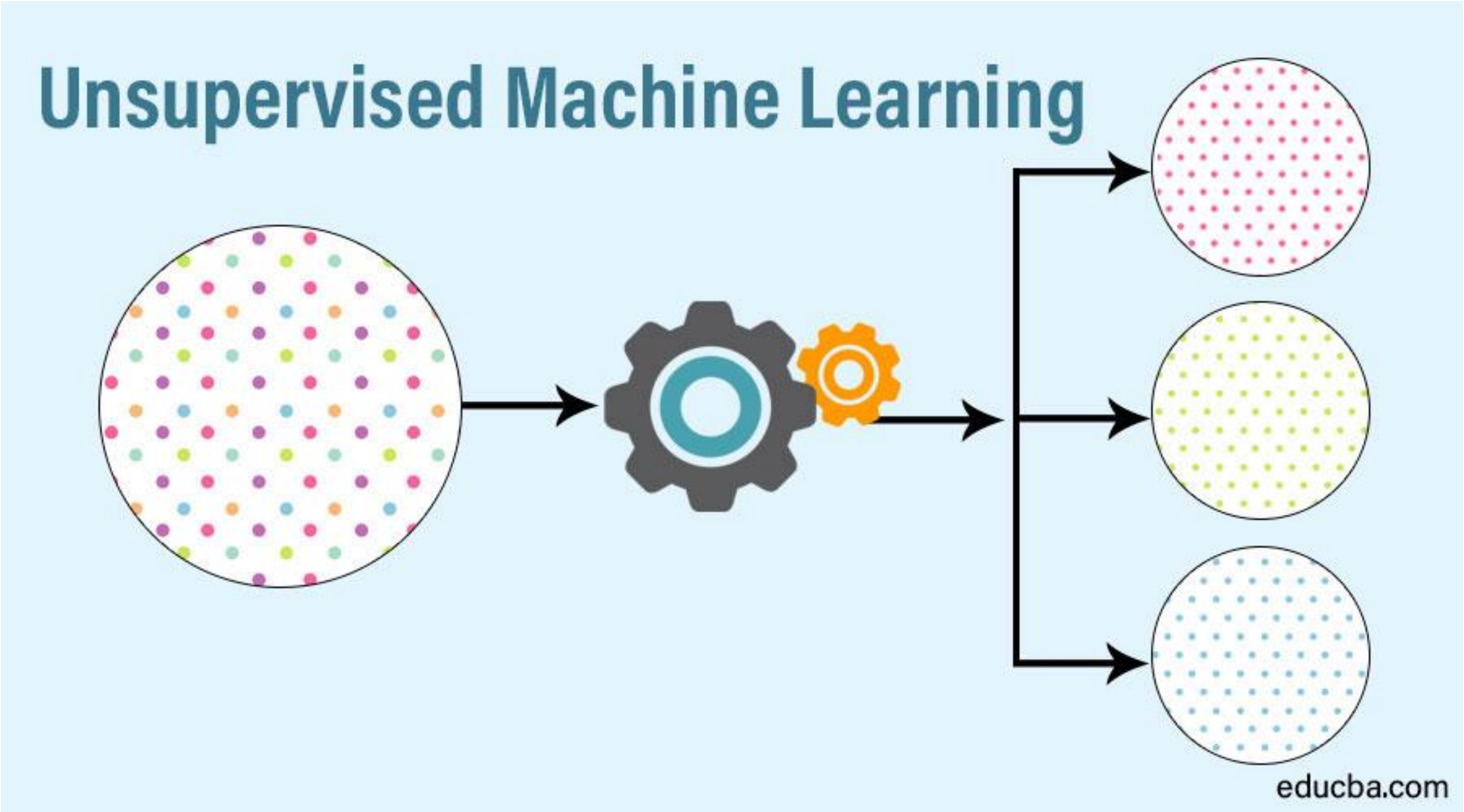
Classification

- Determining whether an email is spam or not.
- Determining whether a given image portrays a cat or a dog
- Categorizing videos on YouTube.

Applications of Supervised Learning

- BioInformatics
- Speech Recognition
- Spam Detection
- Object Recognition for vision

Unsupervised Learning

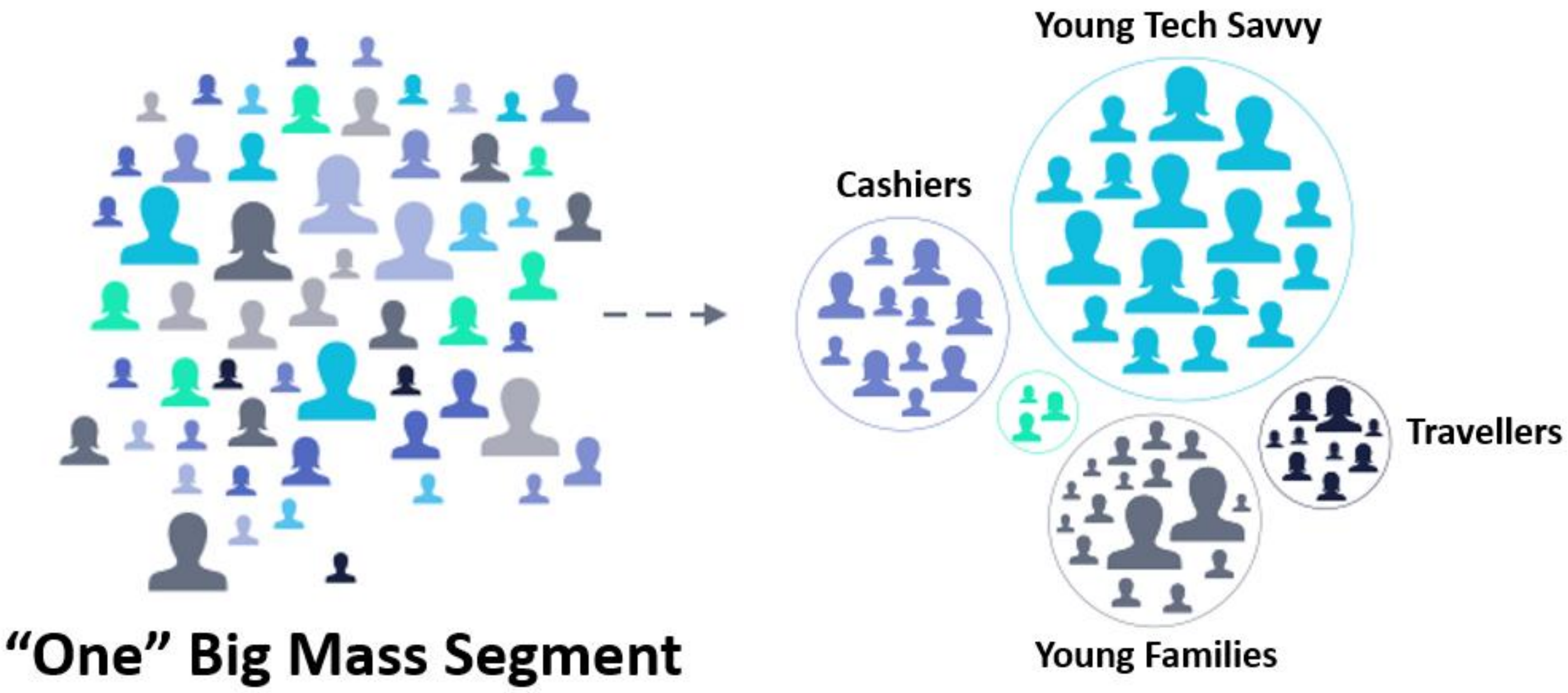


Unsupervised Learning

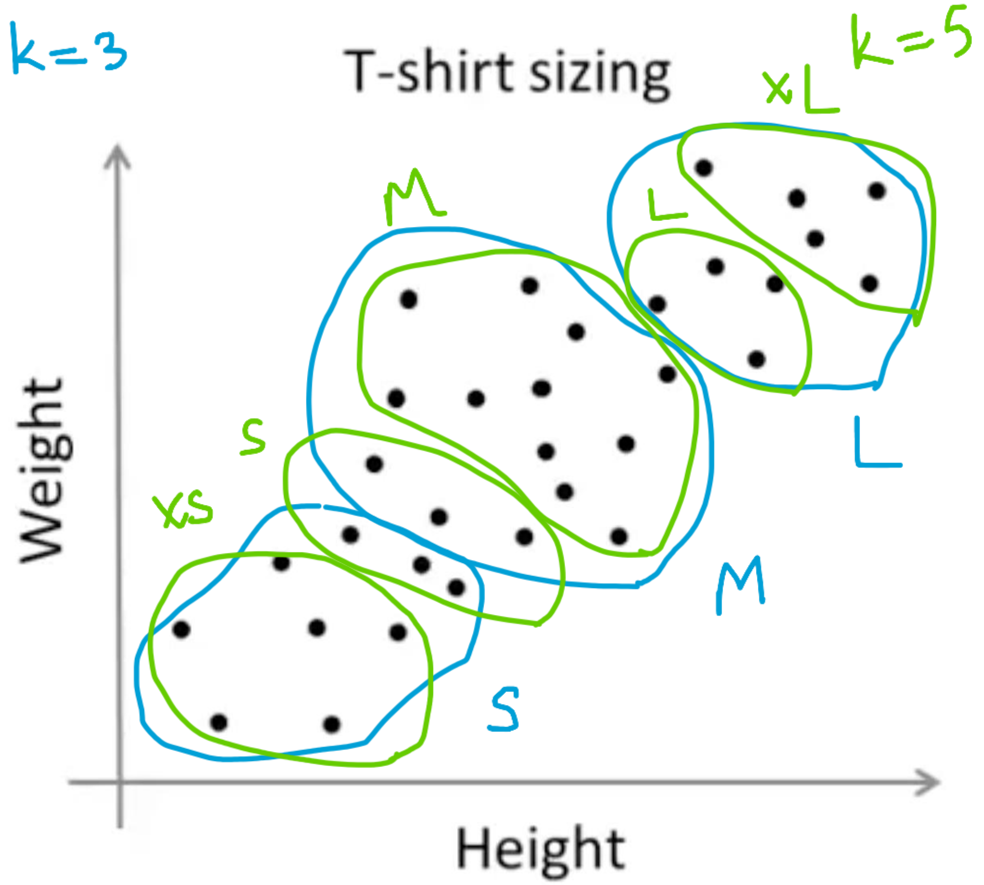
Clustering –has no pre-defined classes. Group data points into “clusters”, try to find structure in the data.

- k-means

Unsupervised Learning



Unsupervised Learning

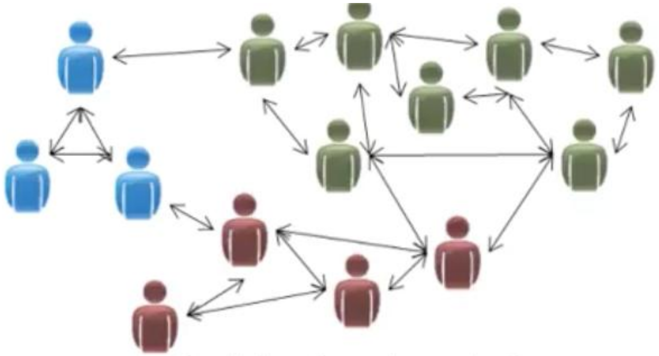


Of the following examples, which would you address using an <u>unsupervised</u> learning algorithm? (Check all that apply.)

- A Given email labeled as spam/not spam, learn a spam filter.
- B Given a set of news articles found on the web, group them into set of articles about the same story.
- C Given a database of customer data, automatically discover market segments and group customers into different market segments.
- D Given a dataset of patients diagnosed as either having diabetes or not, learn to classify new patients as having diabetes or not.

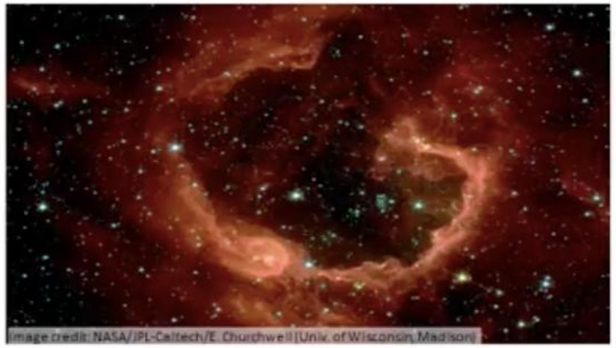
Applications of Unsupervised Learning

Organize computing clusters



Social network analysis

Market segmentation



Astronomical data analysis

- **Grocery shop or e-commerce store/ marketplace**
- **Social Media Platform**
- **Services**
- **Banking**
- **Politics**
- **Data Visualization**
- **Entertainment**
- **Image segmentation**
- **Content**
- **Structural discovery**

Machine Learning Life Cycle

1. Identify the problem – Business Understanding
2. Data Preparation – collecting and cleaning
3. Choosing machine learning algorithm
4. Training the model
5. Testing the model
6. Model Deployment

Model building

Machine Learning Life Cycle

- 1. Identify the problem – Business Understanding
    - Determine
    - Understand
    - Mapping

Machine Learning Life Cycle

- 2. Data Preparation – collecting and cleaning
    - Identify
    - Collect
    - Process

Machine Learning Life Cycle

- 2. Data Preparation – collecting and cleaning

- What if we could predict the occurrence of diabetes and appropriate measures beforehand to prevent it?

Attributes:

1.npreg – Number of times pregnant
2.glucose – Plasma glucose concentration
3.bp – Blood pressure
4.skin – Triceps skinfold thickness
5.bmi – Body mass index
6.ped – Diabetes pedigree function
7.age – Age
8.income – Income

;npreg;glu;bp;skin;bmi;ped;age,income
1;6;148;72;35;33.6;0.627;50
2;1;85;66;29;26.6;0.351;31
3;1;89;80;23;28.1;0.167;21
4;3;78;50;32;31;0.248;26
5;2;197;70;45;30.5;0.158;53
6;5;166;72;19;25.8;0.587;51
7;0;118;84;47;45.8;0.551;31
8;1;103;30;38;43.3;0.183;33
9;3;126;88;41;39.3;0.704;27
10;9;119;80;35;29;0.263;29
11;1;97;66;15;23.2;0.487;22
12;5;109;75;26;36;0.546;60
13;3;88;58;11;24.8;0.267;22
14;10;122;78;31;27.6;0.512;45
15;4;97;60;33;24;0.966;33
16;9;102;76;37;32.9;0.665;46
17;2;90;68;42;38.2;0.503;27
18;4;111;72;47;37.1;1.39;56
19;3;180;64;25;34;0.271;26
20;7;106;92;18;39;0.235;48
21;9;171;110;24;45.4;0.721;54

Machine Learning Life Cycle

- 2. Data Preparation – collecting and cleaning

	npreg	glu	bp	skin	bmi	ped	age	income
1	6	148	72	35	33.6	0.627	50	
2	1	85	66	29	26.6	0.351	31	
3	1	89	6600	23	28.1	0.167	21	
4	3	78	50	32	31	0.248	26	
5	2	197	70	45	30.5	0.158	53	
6	5	166	72	19	25.8	0.587	51	
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15	4		60	33	24	0.966	33	
16	9	102	76	37	32.9	0.665	46	
17	2	90	68	42	38.2	0.503	27	
18	4	111	72	47	37.1	1.39	56	
19	3	180	64	25	34	0.271	26	
20	7	106	92	18		0.235	48	
21	9	171	110	24	45.4	0.721	54	

	npreg	glu	bp	skin	bmi	ped	age
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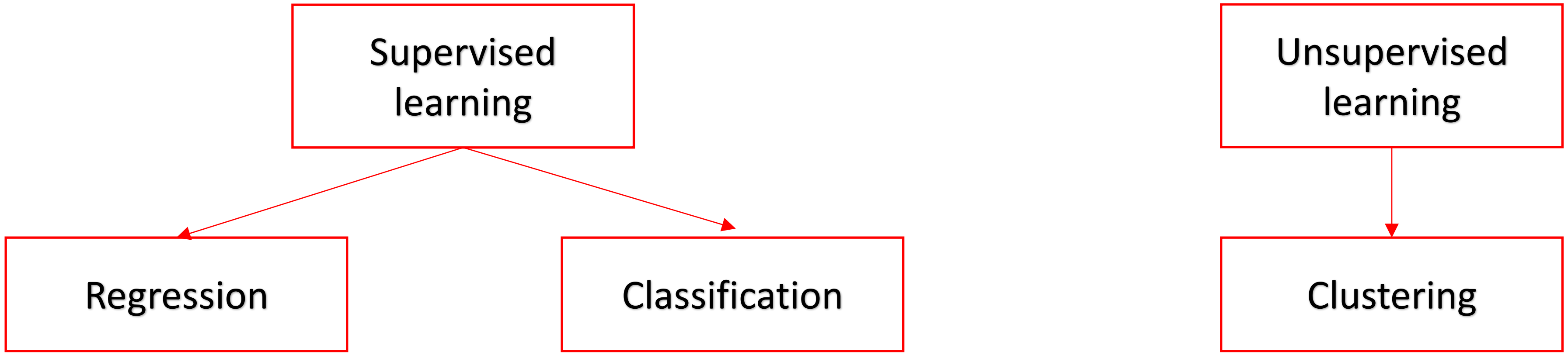
20	7	106	92	18	39	0.235	48
21	9	171	110	24	45.4	0.721	54

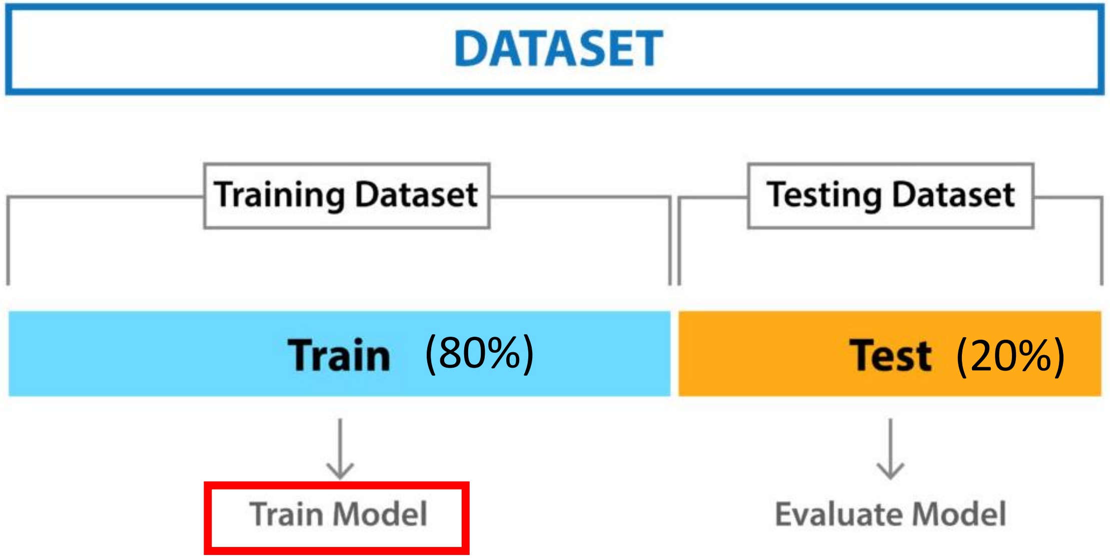
Machine Learning Lifecycle

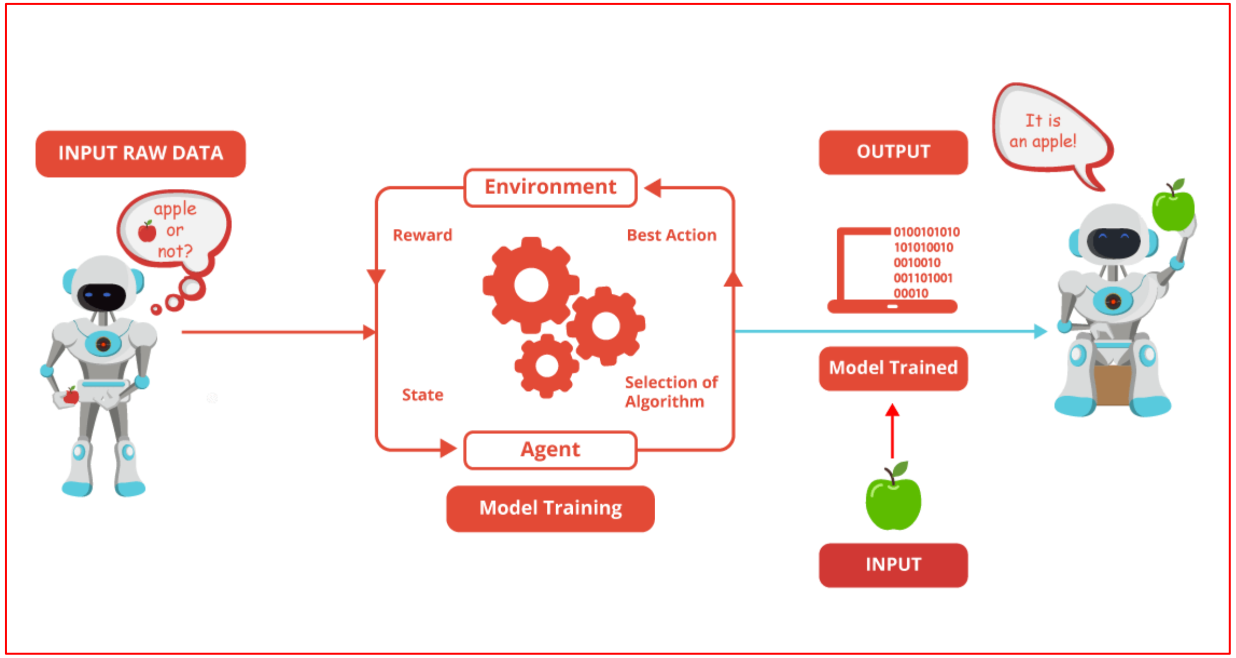
3. Choosing machine learning algorithm



Machine Learning Life Cycle

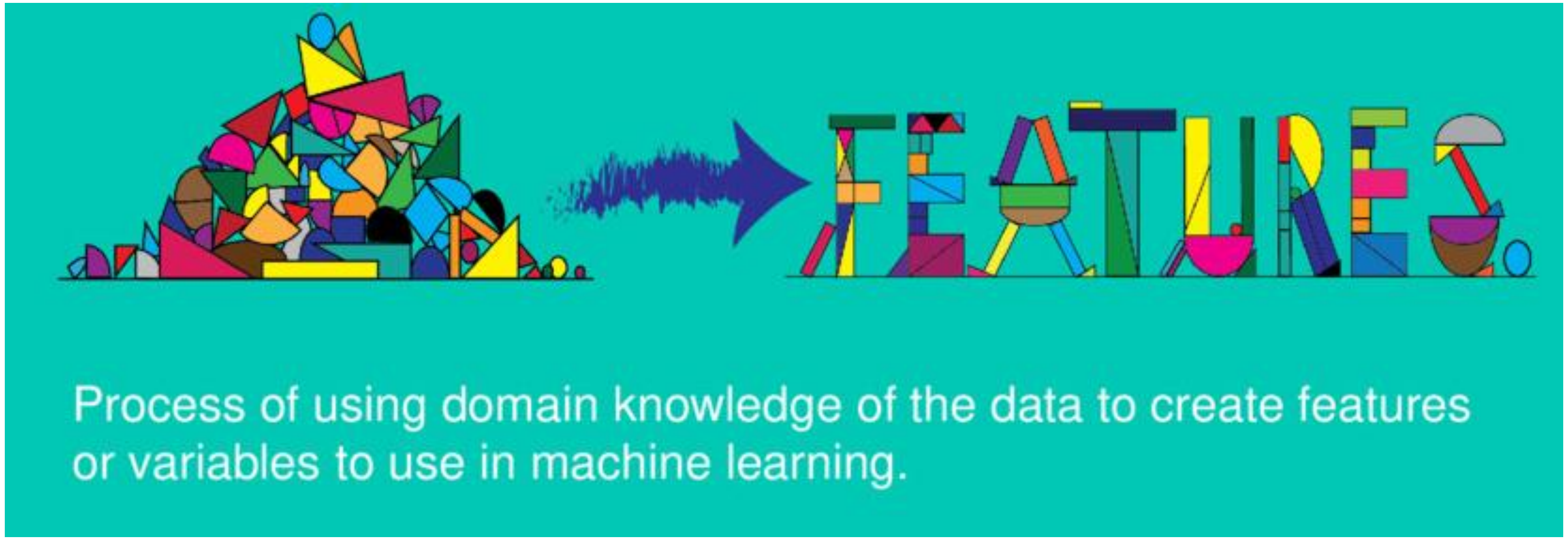
4. Training the model





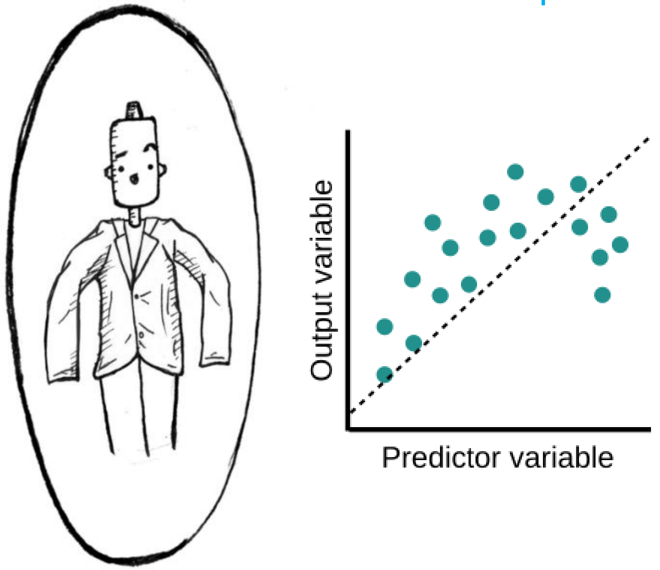
Feature Engineering

- Critical to correclty predict the target value
- More than feature selection
- Improve the performance of the machine learning models

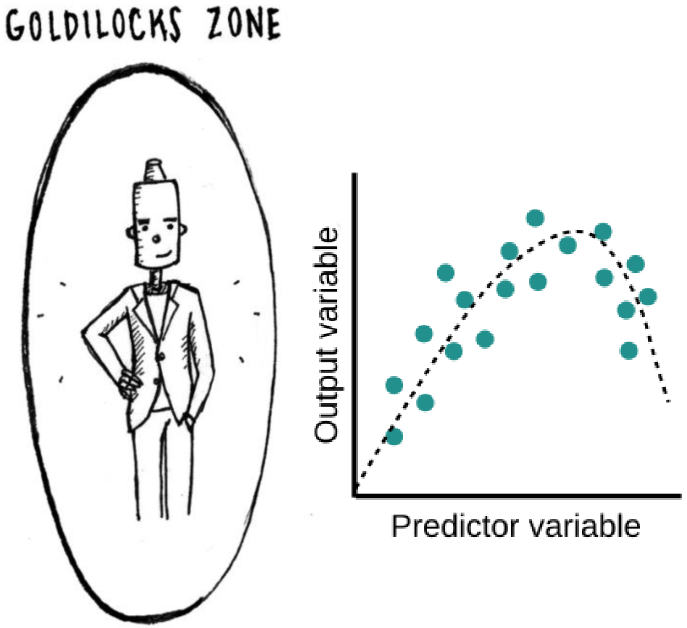


Underfitting vs Overfitting (and Best Fitting)

Learning too little of the true concept

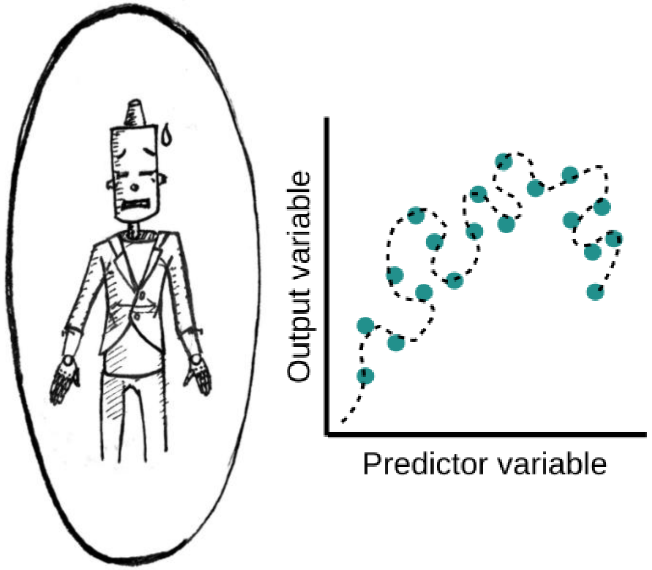


High bias error
and low variance



Balance between
bias and variance

Fitting the data too well



High variance error
and low bias

Overfitting and Underfitting

Techniques to reduce underfitting:

- Increase model complexity
- Increase the number of features, performing feature engineering
- Remove noise from the data.
- Increase the duration of training

Techniques to reduce overfitting:

- Increase training data
- Reduce model complexity
- Early stopping
- Regularization
- Use dropout

Machine Learning Life Cycle

4. Testing the model

- Confusion Matrix
  - $Accuracy = \frac{TP+TN}{TP+FP+FN+TN}$
  - $Precision = \frac{TP}{TP+FP}$
  - $Recall = \frac{TP}{TP+FN}$
  - $F1\ score = \frac{2TP}{2TP+FP+FN}$

		Actual	
		Positive	Negative
Predicted	Positive	True Positive (TP)	False Positive (FP) **Type 1 error**
	Negative	False Negative (FN) **Type 2 error**	True Negative (TN)

Machine Learning Life Cycle

4. Testing the model

- Mean absolute error

$$\text{MAE} = \frac{1}{n}\sum_{t=1}^{n} |e_t|$$

- Mean squared error

$$\text{MSE} = \frac{1}{n}\sum_{t=1}^{n} e_t^2$$

- Root mean squared error

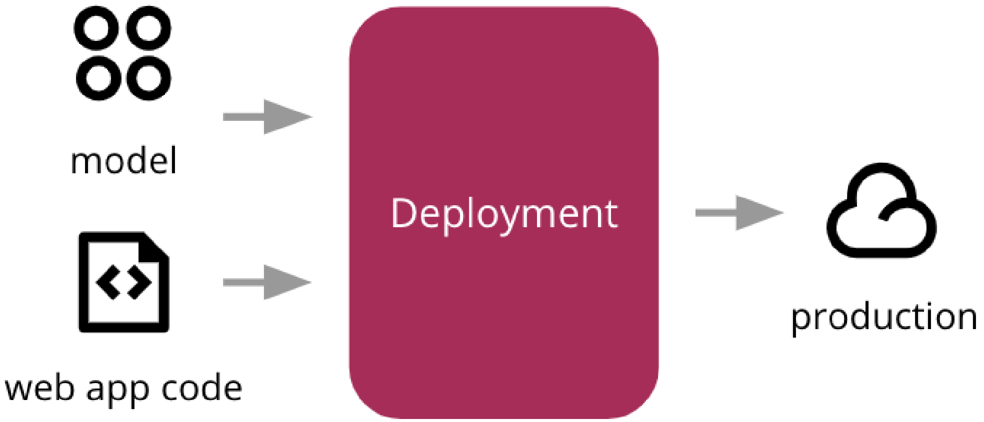
$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{t=1}^{n} e_t^2}$$

- Mean absolute percentage error

$$\text{MAPE} = \frac{100\%}{n}\sum_{t=1}^{n} \left|\frac{e_t}{y_t}\right|$$

Machine Learning Life Cycle

Model Deployment/Operationalize the model



Machine Learning is Everywhere?